Eurobarometer Data Lab

Exploring countries, time, groups, and individuals in R

[Your Names]

2026-10-16

Getting started

This file is a toolbox for your own exploration of Eurobarometer survey data. Run an example, then change the countries, survey waves, groups or questions. The aim is to discover the different questions we can investigate with individual responses: Who thinks what? Do groups differ? Do country patterns change over time? Are two attitudes related within the same people?

Work in pairs. Spend about **40 minutes exploring** and keep the final **10 minutes for discussion**. Inspect the data in section 1, then choose **two of the four examples in section 2**. You do not need to complete them all. Finish by developing a research question in section 3.

In RStudio, use the green arrow to run a code chunk. In Positron, use **Run Cell**. **Render** in RStudio or **Preview** in Positron runs the whole document. Write your observations in the spaces below.

Preparation

Download the data

We use the **Mannheim Eurobarometer Trend File 1970–2002 (ZA3521), version 2.0.1**. This historical dataset combines harmonized questions across waves, so we can explore countries, time and individuals with one download. It does not describe current public opinion.

Each row here represents **one respondent in one survey wave**. Eurobarometer draws new samples across waves: a time comparison describes population patterns, not changes in the same people.

1. Create a free GESIS account and sign in to the GESIS study page for ZA3521.
2. Download the **SPSS data file** and unzip it if necessary. Also open the codebook.
3. Put the `.sav` file in a folder named `data` next to this `.qmd` file. Rename your local copy to `ZA3521.sav`, or change `data_file` below.

Use the study's download and citation information and follow its terms of use. For a smooth start, download the file and install the packages before class.

Install packages once & Working directory

Run this chunk manually. It is excluded from rendering (`eval: false`). Installation requires an internet connection.

Ideally, you store both your data and the .qmd R-Script in one folder. Only if you tell RStudio/ Positron your working directory, can it find the required data.

```
install.packages(c("tidyverse", "haven", "rio"))

getwd() # get's your current working directory

setwd("/your-directory/folder ") # to set to "your-directory"
# for example, mine is:
setwd("/Users/DATA/_GIT/UE-Data-in-CP/Eurobarometer Data Lab")
```

Load packages and data

```
library(tidyverse)
library(haven)

data_file <- "data/ZA3521.sav"
if (!file.exists(data_file)) {
  stop("Download ZA3521 from GESIS and put the SPSS file at: ", data_file,
       ". See Preparation above, or change data_file.")
}

eb_raw <- read_sav(data_file)
# Use lower-case column names throughout this lab.
names(eb_raw) <- tolower(names(eb_raw))
```

1. Inspect the data and find a question

Look at respondents and variable labels

```
dim(eb_raw)
```

```
[1] 1134384     146
```

```
eb_raw |> select(eb, year, nation2, satislfe, membrshp, age, lrs) |> head()
```

```
# A tibble: 6 × 7
  eb         year nation2    satislfe  membrshp  age       lrs
```

```
  <dbl+lbl> <dbl> <dbl+lbl>  <dbl+lbl> <dbl+lbl> <dbl+lbl> <dbl+lbl>
1 1 [ECS70]  1970 1 [FRANCE] NA        NA        NA        NA
2 1 [ECS70]  1970 1 [FRANCE] NA        NA        NA        NA
3 1 [ECS70]  1970 1 [FRANCE] NA        NA        NA        NA
4 1 [ECS70]  1970 1 [FRANCE] NA        NA        NA        NA
5 1 [ECS70]  1970 1 [FRANCE] NA        NA        NA        NA
6 1 [ECS70]  1970 1 [FRANCE] NA        NA        NA        NA
```

```
# SPSS stores descriptions alongside the short variable names.
variable_guide <- tibble(
  variable = names(eb_raw),
  description = map_chr(eb_raw, function(x) {
    label <- attr(x, "label", exact = TRUE)
    if (is.null(label)) "" else as.character(label)
  })
)

variable_guide |>
  filter(str_detect(description,
                    regex("satisf|trust|education|occupation|political",
                          ignore_case = TRUE)))
```

```
# A tibble: 22 × 2
   variable description
   <chr>    <chr>
 1 trustep  TRUST IN EURO PARLIAMENT
 2 trustec  TRUST IN EURO COMMISSION
 3 trustcm  TRUST IN COUNCIL OF MINISTERS OF 'EU'
 4 trustcj  TRUST IN COURT OF JUSTICE OF 'EC'
 5 trusteo  TRUST IN EURO OMBUDSMAN
 6 trustecb TRUST IN EURO CENTRAL BANK
 7 trusteca TRUST IN EURO COURT OF AUDITORS
 8 trustcr  TRUST IN COMM OF REGIONS OF 'EU'
 9 trustsec TRUST IN SOCIAL AND ECON COMM OF 'EU'
10 satislfe SATISFACTION LIFE
# ℹ 12 more rows
```

```
# Response scales, codes and their labels for two example questions
attr(eb_raw$satislfe, "labels")
```

```
  VERY SATISFIED FAIRLY SATISFIED    NOT VERY SAT.  NOT AT ALL SAT.
               1                2                3               4
          DK,NA             INAP
               8                9
```

```
attr(eb_raw$membrshp, "labels")
```

```
 GOOD THING NEITHER NOR   BAD THING       DK,NA        INAP
          1           2           3           8           9
```

```
attr(eb_raw$lrs, "labels")
```

```
   LEFT       2       3       4       5       6       7       8       9   RIGHT
      1       2       3       4       5       6       7       8       9      10
REFUSED   DK,NA    INAP
     96      98      99
```

For a spreadsheet-style view, run these commands manually:

```
View(variable_guide)
View(eb_raw)
```

Find a question that interests you in the codebook. Check its wording, response options, countries and waves. The descriptions stored in the data are a starting point; the codebook also documents changes and exceptions.

The examples use `satislfe` (life satisfaction), `membrshp` (evaluation of EC/EU membership), `age`, and `lrs` (left–right self-placement). See codebook pages **167, 64, 296 and 249**, respectively.

Your notes: What would you like to know about public opinion? Which question could help you investigate it? What does it leave out?

Write here.

Prepare a small set of variables

Survey files contain numerical codes that are not substantive answers. We keep valid responses and turn other codes into `NA` (missing). We retain respondents with missing answers so that we can check how many are excluded from each comparison.

```
eb_data <- eb_raw |>
  transmute(
    wave = as.integer(eb),
```

```
    year = as.integer(year),
    country_id = as.integer(nation2),
    country = as_factor(nation2),
    weight = as.numeric(wnation),
    life_satisfaction = if_else(satislfe %in% 1:4,
                                as.numeric(satislfe), NA_real_),
    membership = if_else(membrshp %in% 1:3,
                         as.numeric(membrshp), NA_real_),
    age = if_else(between(as.numeric(age), 15, 99),
                  as.numeric(age), NA_real_),
    left_right = if_else(lrs %in% 1:10,
                         as.numeric(lrs), NA_real_)
  ) |>
  mutate(
    # 1 = satisfied; 0 = dissatisfied; missing stays missing.
    satisfied = if_else(is.na(life_satisfaction), NA_real_,
                        as.numeric(life_satisfaction <= 2)),
    eu_positive = if_else(is.na(membership), NA_real_,
                          as.numeric(membership == 1)),
    age_group = cut(age, breaks = c(15, 30, 50, 65, Inf),
                    labels = c("15–29", "30–49", "50–64", "65+"),
                    right = FALSE)
  )
```

`satisfied` combines the two satisfied categories; `eu_positive` identifies a favourable membership evaluation. The latter compares this answer with all other substantive answers, including the neutral category. Both exclude nonresponses. Collapsing categories makes a graph simple but loses detail.

The examples use **national weights** (`wnation`) with `nation2`, which combines the German and UK subsamples (codebook pages 42 and 59). All percentages below are calculated separately within a country and wave. An EU-wide population estimate would require a different weighting choice. `n_valid` always counts actual respondents, not weighted population totals.

Choose your cases

```
# Discover available countries, waves and years.
eb_data |> distinct(country_id, country) |> arrange(country_id)
```

```
# A tibble: 16 × 2
   country_id country
        <int> <fct>
 1          1 FRANCE
```

```
 2          2 BELGIUM
 3          3 NETHERLANDS
 4          4 GERMANY
 5          5 ITALY
 6          6 LUXEMBOURG
 7          7 DENMARK
 8          8 IRELAND
 9          9 UK
10         11 GREECE
11         12 SPAIN
12         13 PORTUGAL
13         15 NORWAY
14         16 FINLAND
15         17 SWEDEN
16         18 AUSTRIA
```

```
eb_data |> distinct(wave, year) |> arrange(desc(wave)) |> head(15)
```

```
# A tibble: 15 × 2
    wave  year
   <int> <int>
 1   572  2002
 2   571  2002
 3   570  2002
 4   563  2002
 5   562  2001
 6   561  2001
 7   560  2001
 8   552  2001
 9   551  2001
10   550  2001
11   542  2001
12   541  2000
13   540  2000
14   530  2000
15   521  1999
```

```
range(eb_data$year)
```

```
[1] 1970 2002
```

```
# Country codes: France = 1, Germany = 4, Denmark = 7, UK = 9.
# Look at the country table above to choose others.
chosen_countries <- c(1, 4, 7)
chosen_country <- 4
chosen_wave <- 571  # EB 57.1, spring 2002; see codebook p. 27.
first_year <- 1991
last_year <- 2002

one_wave <- eb_data |>
  filter(wave == chosen_wave, country_id %in% chosen_countries)

# Coverage before dropping missing answers
one_wave |>
  group_by(country) |>
  summarise(n_respondents = n(),
            missing_satisfaction = sum(is.na(satisfied)),
            missing_membership = sum(is.na(eu_positive)),
            missing_age = sum(is.na(age)),
            missing_left_right = sum(is.na(left_right)),
            unusable_weights = sum(is.na(weight) | weight <= 0),
            .groups = "drop")
```

```
# A tibble: 3 × 7
  country n_respondents missing_satisfaction missing_membership missing_age
  <fct>           <int>                <int>              <int>       <int>
1 FRANCE           1010                   13                 48           0
2 GERMANY          2051                   16                166           0
3 DENMARK          1000                    3                 28           0
# ℹ 2 more variables: missing_left_right <int>, unusable_weights <int>
```

Change these settings, then rerun this chunk and your chosen examples. For examples B and C, include `chosen_country` in `chosen_countries`. `filter()` selects rows; `select()` selects columns; `|>` passes a result to the next step. `%in%` matches any value in a list.

2A. Compare countries

Question: Does life satisfaction differ between countries in the same wave?

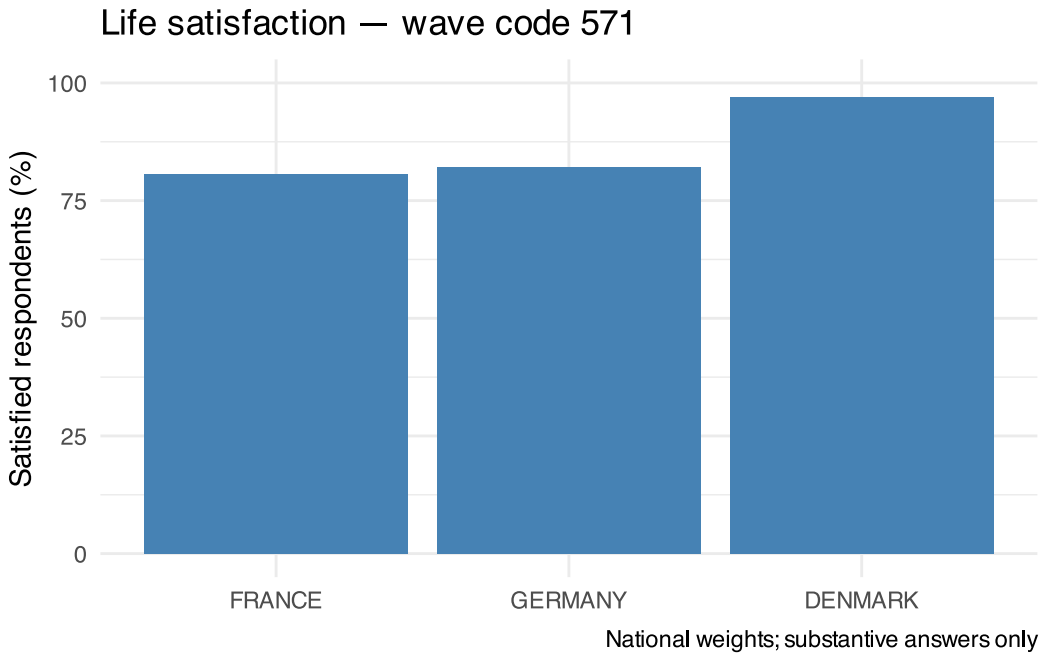
```
country_results <- one_wave |>
  filter(!is.na(satisfied), !is.na(weight), weight > 0) |>
  group_by(country, year, wave) |>
  summarise(n_valid = n(),
            percent_satisfied = 100 * weighted.mean(satisfied, weight),
```

```
            .groups = "drop")

country_results
```

```
# A tibble: 3 × 5
  country  year  wave n_valid percent_satisfied
  <fct>   <int> <int>   <int>             <dbl>
1 FRANCE   2002   571     997              80.7
2 GERMANY  2002   571    2035              82.1
3 DENMARK  2002   571     997              96.9
```

```
country_plot <- ggplot(country_results,
                       aes(x = country, y = percent_satisfied)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = paste("Life satisfaction — wave code", chosen_wave),
       x = NULL, y = "Satisfied respondents (%)",
       caption = "National weights; substantive answers only") +
  theme_minimal()
country_plot
```



Try it: Add another country. Or replace `satisfied` in the filter and weighted mean with `eu_positive` and update the column and graph labels.

Your notes: Describe one difference. Who is included in the percentage? Could a country average conceal differences between people within it?

Write here.

2B. Compare groups within a country

Question: Are younger and older respondents equally positive about EC/EU membership?

```
# get the age groups available in the age_group variable for your chosen country
one_wave$age_group[one_wave$wave   ==   chosen_wave   &   one_wave$country   ==
chosen_country]
```

```
factor()
Levels: 15–29 30–49 50–64 65+
```

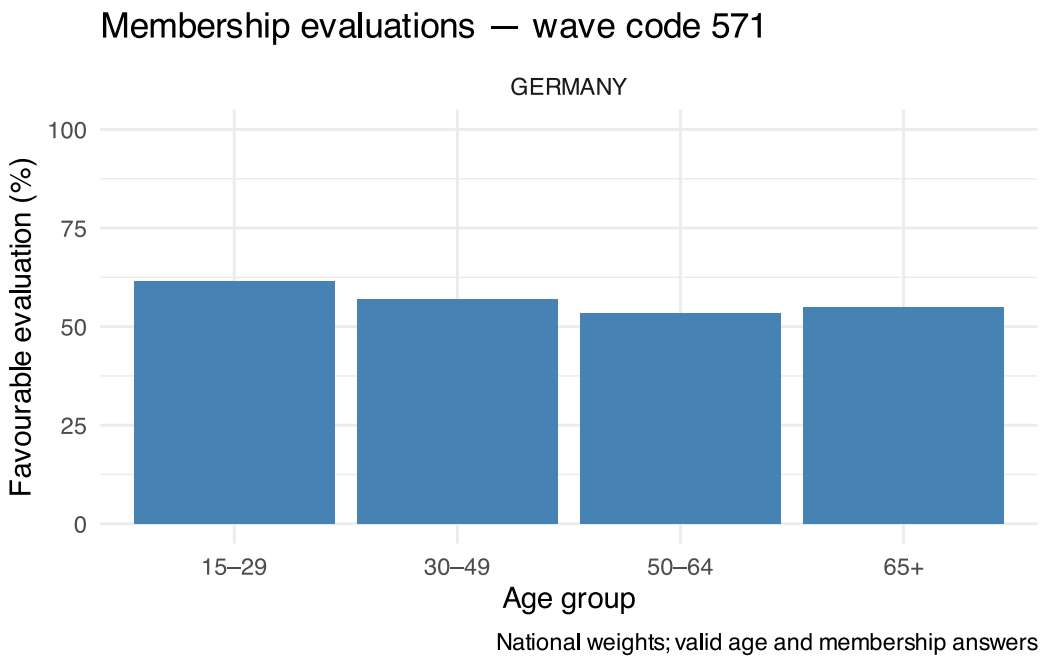
```
group_results <- one_wave |>
  filter(country_id == chosen_country,
         !is.na(age_group), !is.na(eu_positive),
         !is.na(weight), weight > 0) |>
  group_by(country, age_group) |>
  summarise(n_valid = n(),
            percent_positive = 100 * weighted.mean(eu_positive, weight),
            .groups = "drop")

group_results
```

```
# A tibble: 4 × 4
  country age_group n_valid percent_positive
  <fct>   <fct>       <int>            <dbl>
1 GERMANY 15–29         324             61.4
2 GERMANY 30–49         691             57.0
3 GERMANY 50–64         497             53.5
4 GERMANY 65+           373             54.9
```

```
group_plot <- ggplot(group_results,
                     aes(x = age_group, y = percent_positive)) +
  geom_col(fill = "steelblue") +
  facet_wrap(~ country) +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = paste("Membership evaluations — wave code", chosen_wave),
       x = "Age group", y = "Favourable evaluation (%)",
```

```
       caption = "National weights; valid age and membership answers") +
  theme_minimal()
group_plot
```



Try it: Remove `country_id == chosen_country` from the filter to compare the age pattern across all your selected countries. Or change the age-group boundaries in section 1 and rerun the preparation and this example.

For another grouping, use the variable guide to find education, occupation or another demographic variable. Check its codes before adding it to `eb_data`. An age difference in one wave does not tell us whether people change their opinions as they get older.

Your notes: Which groups differ? Check their sample sizes. Does changing the group boundaries change your impression?

Write here.

2C. Connect two answers from the same individual

Question: Is political self-placement related to membership evaluations within a country?

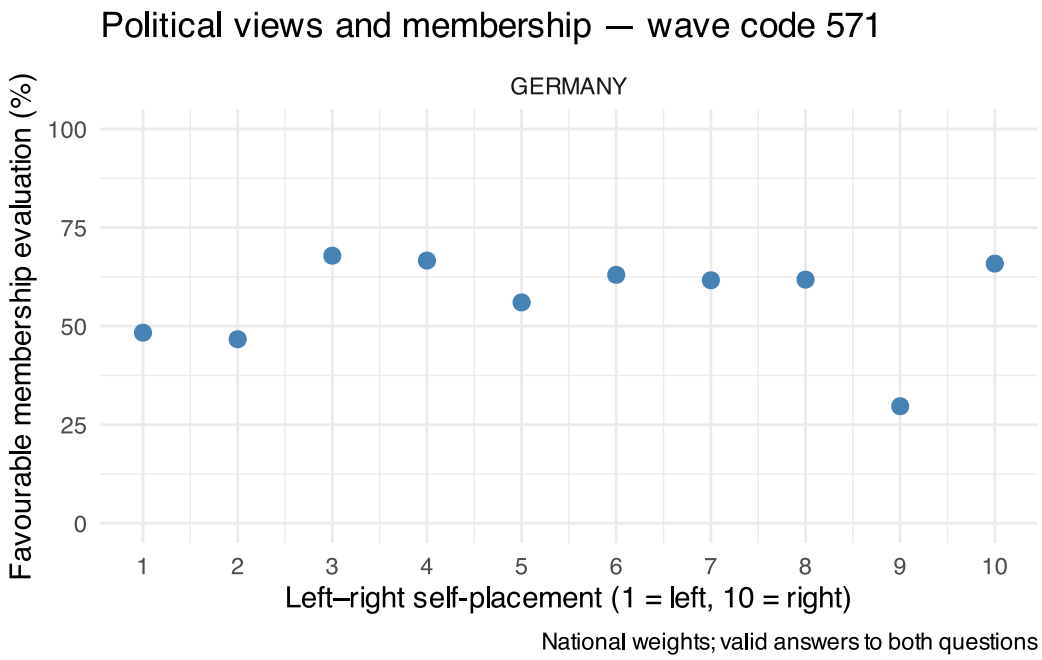
We connect two answers given by the **same respondents**. This is different from comparing country averages. A relationship between national averages does not necessarily hold between individuals.

```r
individual_results <- one_wave |>
  filter(country_id == chosen_country,
         !is.na(left_right), !is.na(eu_positive),
         !is.na(weight), weight > 0) |>
  group_by(country, left_right) |>
   summarise(n_valid = n(), # returns the number of individuals per scale-step
on left-right-selfplacement
             percent_positive = 100 * weighted.mean(eu_positive, weight), #
returns % favorable membership within each scale-step
            .groups = "drop")

individual_results
```

```
# A tibble: 10 × 4
   country left_right n_valid percent_positive
   <fct>        <dbl>   <int>            <dbl>
 1 GERMANY          1      55             48.3
 2 GERMANY          2      60             46.7
 3 GERMANY          3     160             67.9
 4 GERMANY          4     202             66.6
 5 GERMANY          5     468             56.0
 6 GERMANY          6     264             63.0
 7 GERMANY          7     164             61.6
 8 GERMANY          8      89             61.8
 9 GERMANY          9      34             29.7
10 GERMANY         10      36             65.9
```

```r
individual_plot <- ggplot(individual_results,
                          aes(x = left_right, y = percent_positive)) +
  geom_point(size = 2.5, colour = "steelblue") +
  facet_wrap(~ country) +
  scale_x_continuous(breaks = 1:10) +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = paste("Political views and membership – wave code", chosen_wave),
       x = "Left–right self-placement (1 = left, 10 = right)",
       y = "Favourable membership evaluation (%)",
       caption = "National weights; valid answers to both questions") +
  theme_minimal()
individual_plot
```



Each dot summarises respondents at one scale position. It is not a single person. Consult `n_valid`: a dot based on few answers may be unstable.

Try it: Replace `eu_positive` in the filter and weighted mean with `satisfied`; update the labels. Or remove the country filter to see whether the relationship looks similar in other countries.

Your notes: Is there a pattern? Is it linear, curved or unclear? Suggest one other characteristic that could help explain the association. This comparison alone does not establish that either attitude causes the other.

Write here.

2D. Compare countries over time

Question: Do countries follow similar paths in life satisfaction?

First calculate each wave separately. For a simple calendar-year graph, then keep the last wave with valid satisfaction answers in each country and year. This avoids pooling several surveys into one estimate.

```
time_results <- eb_data |>
  filter(country_id %in% chosen_countries,
         year >= first_year, year <= last_year) |>
  group_by(country, wave, year) |>
  summarise(
    n_valid = sum(!is.na(satisfied) & !is.na(weight) & weight > 0),
    percent_satisfied = if (n_valid > 0) {
```

```
      valid <- !is.na(satisfied) & !is.na(weight) & weight > 0
      100 * weighted.mean(satisfied[valid], weight[valid])
    } else NA_real_,
    .groups = "drop"
  )

# Coverage and nonresponse vary across waves.
time_results |> arrange(country, wave) |> head(20)
```

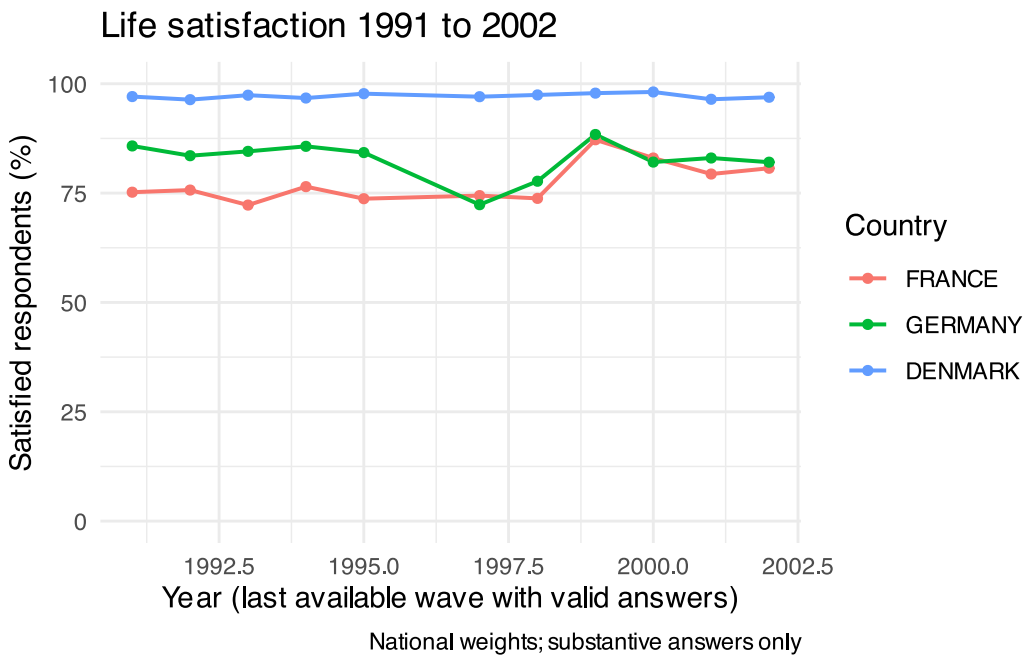
```
# A tibble: 20 × 5
   country  wave  year n_valid percent_satisfied
   <fct>   <int> <int>   <int>             <dbl>
 1 FRANCE    350  1991     990              80.7
 2 FRANCE    351  1991       0              NA
 3 FRANCE    360  1991     996              75.2
 4 FRANCE    370  1992     994              71.6
 5 FRANCE    371  1992    1002              79.2
 6 FRANCE    380  1992     996              74.9
 7 FRANCE    381  1992    1002              75.7
 8 FRANCE    390  1993    1016              69.7
 9 FRANCE    391  1993       0              NA
10 FRANCE    400  1993    1022              72.3
11 FRANCE    410  1994     993              73.4
12 FRANCE    411  1994       0              NA
13 FRANCE    420  1994    1004              76.5
14 FRANCE    430  1995       0              NA
15 FRANCE    431  1995     991              73.7
16 FRANCE    432  1995       0              NA
17 FRANCE    440  1995       0              NA
18 FRANCE    441  1995       0              NA
19 FRANCE    451  1996       0              NA
20 FRANCE    460  1996       0              NA
```

```
annual_results <- time_results |>
  filter(n_valid > 0) |>
  group_by(country, year) |>
  slice_max(wave, n = 1, with_ties = FALSE) |>
  ungroup()

# Check which wave represents each country-year.
annual_results |> select(country, year, wave, n_valid)
```

```
# A tibble: 33 × 4
   country  year  wave n_valid
   <fct>   <int> <int>   <int>
 1 FRANCE   1991   360     996
 2 FRANCE   1992   381    1002
 3 FRANCE   1993   400    1022
 4 FRANCE   1994   420    1004
 5 FRANCE   1995   431     991
 6 FRANCE   1997   471     998
 7 FRANCE   1998   490    1020
 8 FRANCE   1999   521    1002
 9 FRANCE   2000   541     994
10 FRANCE   2001   562     988
# ℹ 23 more rows
```

```
time_plot <- ggplot(annual_results,
                    aes(x = year, y = percent_satisfied,
                        colour = country, group = country)) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 1.3) +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = paste("Life satisfaction", first_year, "to", last_year),
       x = "Year (last available wave with valid answers)",
       y = "Satisfied respondents (%)", colour = "Country",
       caption = "National weights; substantive answers only") +
  theme_minimal()
time_plot
```



Use the wave column and the fieldwork table on codebook page 27 to check when within each year the selected survey took place. Compare the same wave across countries when possible. Years without valid answers are absent; missing data do not imply zero satisfaction. Lines guide the eye and do not establish what happened between observations.

Try it: Change the period or countries. Alternatively, use `eu_positive` instead of `satisfied` throughout this chunk and update the labels. Check whether the two questions were asked in the same waves.

Your notes: Identify one change or stable pattern. What could explain it? What additional evidence would help? Samples and question coverage can change. Germany before reunification covers a different territory; the default period starts in 1991 to avoid that break.

Write here.

3. Develop a research question

Bring your two explorations together. Write down:

1. One observation and one research question it suggests.
2. Whether your question concerns countries, change over time, groups or relationships between individuals' characteristics and attitudes.
3. The countries, waves, population and variables you would investigate.
4. One comparison, additional variable or data source you would need next.
5. Something your current graphs cannot establish.

For discussion, prepare **one graph and two sentences**: what it shows and what you would investigate next. A surprising result or a difficulty with measurement is also useful.

Write here.

The graphs are descriptive estimates. We have not calculated sampling uncertainty or accounted for the full sampling design in standard errors. Treat small differences cautiously, especially in small groups. Weighting does not remove all sources of survey error.

Optional tools: Keep more detail and save your work

The examples reduce satisfaction to two categories. Inspect the original four categories to see what that simplification hides. These are **unweighted counts**, useful for checking the data rather than estimating population shares.

```
one_wave |>
  count(country, life_satisfaction, .drop = FALSE)
```

```
# A tibble: 29 × 3
   country     life_satisfaction     n
   <fct>                   <dbl> <int>
 1 FRANCE                      1   144
 2 FRANCE                      2   658
 3 FRANCE                      3   151
 4 FRANCE                      4    44
 5 FRANCE                     NA    13
 6 BELGIUM                    NA     0
 7 NETHERLANDS                NA     0
 8 GERMANY                     1   298
 9 GERMANY                     2  1302
10 GERMANY                     3   367
# ℹ 19 more rows
```

Run these commands manually to save one result table and graph:

```
# After running example A; use group_results/group_plot, etc. for other examples.
write_csv(country_results, "data/my-eurobarometer-countries.csv")
ggsave("output/my-eurobarometer-countries.png", plot = country_plot,
       width = 8, height = 5, dpi = 300)
```

Sources and reproducibility

- GESIS study and download: Mannheim Eurobarometer Trend File (ZA3521)
- Dataset DOI: version 2.0.1
- Codebook: variable definitions, coverage and fieldwork dates
- GESIS Eurobarometer FAQ: access, repeated cross-sections and documentation
- Other Eurobarometer trend files

Copy the full dataset citation from GESIS into your work. Record your file version, selected countries and waves, recoding decisions and weight. For a recent survey or another trend file, check its own question wording, variable names, response codes and weights before adapting this script.

HTML is the default output. Select Typst for PDF output.